

Cache Translation Across Heterogeneous LLMs: An Audit with Identity Controls and Oracle Bounds

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Code and all run artifacts: [apcs repository](#) (21 recorded GPU runs)

Abstract

Cache translation maps the key-value (KV) cache that a large teacher model forms over a context into the cache space of a smaller student model, so that the student answers without re-prefilling the context. A growing line of work reports quality-preserving translation across heterogeneous models. We audit this claim on Qwen3 pairs (4B→1.7B and 4B→0.6B) with a three-hypothesis framework and two instruments that prior evaluations omit. First, an *identity control* injects the student’s own cache through the full translation-and-injection pipeline: it reproduces the student’s own prefill exactly (per-sample logit cosine 1.000), which isolates mechanics from mapping. Second, *oracle probes* mix teacher-cache content into the student’s own cache and bound what any translator could extract. Across five mapper families (per-head ridge, affine, per-layer affine, task-aware, and a residual-anchored translator built from the two models’ own projections), a 200-example calibration ladder, and 21 recorded GPU runs, no configuration makes the strong student exceed its own prefill: gold-probability change spans -0.14 to -0.30 against a self-kv control at $+0.000$. The weak student reaches parity ($+0.010$, CI $[-0.08, +0.10]$). Oracle probes show a monotone decline as teacher content replaces student content in early and mid layers, and a perplexity–accuracy decoupling: a translator that restores near-native fluency (PPL 23.7 versus 21.2) still leaves accuracy at 0.267 versus 0.500. We conclude that the teacher’s answer-relevant advantage does not survive KV-space translation under zero re-prefill: it lives in the teacher’s weights, not in its cache. We release the audit protocol as a required checklist for cache-translation claims.

1 Introduction

Multi-model serving systems want to reuse the teacher’s work. A large teacher prefills a long context once; a smaller student should inherit that state and answer at teacher quality without re-reading the context. Cache translation formalizes this idea: learn a map $\mathbf{KV}^T \mapsto \widehat{\mathbf{KV}}^S$ from the teacher’s cache space into the student’s. Recent systems report encouraging results, including Mixture-of-Translators (MoT), which preserves 51.0% closed-set QA accuracy at 7B→0.5B scale [13], Cache-to-Cache semantic communication [5], latent-space communication [3], selective KV sharing [21], and a closed-form ridge mapper that retains 73–98% of a receiver’s standalone-prefill accuracy on four of six matched-KV pairs [10].

These reports conflate three questions that can fail independently. Does the injection pipeline consume a foreign cache correctly (*mechanics*)? Does the map replace the student’s own cache at replacement level (*mapping*)? Does the teacher’s cache carry student-readable content that lifts the student above its own prefill (*exploitability*)? Prior evaluations score the translated end-to-end pipeline against the student, so a positive delta could come from any of the three; a null result cannot be attributed. The line also lacks an identity control, a probe of the exploitability ceiling, and any report of what happens to *accuracy* when perplexity is repaired.

We evaluate cache translation as an audit with three gated hypotheses. **H1 (mechanics)** asks whether a foreign cache is consumed without loss; its instrument is an identity control that injects the student’s own cache through the full pipeline. **H2 (mapping)** asks whether any translator reaches replacement level; its instrument is a ladder of five mapper families under a disjoint calibration ladder of 30 to 200 examples, including a residual-anchored translator (RAT) that we derive from the two models’ own projections and a shared vocabulary alignment. **H3 (exploitability)** asks whether the teacher’s cache contains any student-

readable advantage at all; its instrument is an oracle probe that mixes teacher-cache content into the student’s own cache at controlled fractions and layer windows, which upper bounds every translator that could be trained. Figure 1 shows the gates and the instruments.

The audit returns three verdicts on Qwen3 pairs. H1 passes: the identity control reproduces the student’s own prefill with per-sample logit cosine 1.000 and a gold-probability change of $+0.0001$ (95% CI $[-0.0006, +0.0009]$), so mechanics is not the failure mode. H2 fails for the strong student: across the mapper ladder, the gold-probability change over the student’s own prefill spans -0.138 to -0.302 , and the best configuration reaches 0.365 against a student baseline of 0.503; the weak 0.6B student reaches parity ($+0.010$, CI $[-0.081, +0.102]$). H3 returns a principled negative: with 75% student cache and only 25% teacher content, accuracy already drops from 0.500 to 0.367, the decline is monotone in the teacher fraction, and replacing the top third of layers is the only harmless configuration. A translator that restores near-native fluency (PPL 23.7 versus 21.2 for the identity control) still leaves accuracy at 0.267. The teacher’s advantage does not ride in its cache.

Our contributions are threefold.

1. **An audit framework.** Identity controls, disjoint calibration/evaluation splits, gold-probability metrics with bootstrap intervals and permutation tests, and oracle probes that bound exploitability without training anything (Section 4).
2. **A bounded design space.** Five mapper families, a calibration ladder, component ablations, and a residual-anchored translator derived from the two models’ projections and shared vocabulary, all under one protocol (Section 4 and Section 5).
3. **A verdict with consequences.** Mechanics passes, replacement fails for the strong student and holds for the weak one, and exploitability is bounded at zero; perplexity cannot substitute for accuracy as evidence (Section 5 and Section 7).

2 A Survey of Cache-Level Communication Between Models

We organize prior work into four lines and close each with the assumption it leaves untested. Table 1 summarizes the comparison.

2.1 KV Reuse Within a Single Model

Prompt caching, cache-augmented generation, and streaming attention reuse a model’s own cache across requests. Streaming attention identifies attention sinks at early positions whose KV carries outsized attention mass [22]. CacheGen compresses and streams a KV cache across a network [16]; CacheBlend fuses several cached fragments with selective attention recomputation for RAG [24]; KVSharer shares caches across layers that behave dissimilarly [23]. These systems assume a single model on both sides of the cache. The reuse question is therefore about storage and attention cost, not about representation compatibility. Our audit inherits their serving motivation but moves the mismatch to the model axis, where the compatibility assumption breaks.

2.2 Cross-Model Cache Translation

The line closest to our audit translates caches between models. C2C proposes direct cache-to-cache semantic communication with a depth-ratio layer mapping [5]. LSC aligns caches through a shared latent space with a cross-attention translator [3]. MoT combines gated translator modules over depth-ratio channel windows with a Context Correction Loss that aligns a replayed target trajectory with the native one, and reports the strongest numbers to date: 51.0% average closed-set QA accuracy at Qwen2.5-7B→0.5B and 96.3% retention in long-context cache-augmented generation [13]. A parallel recent system studies cross-model KV cache transfer inside one family under the strict zero-re-prefill regime we audit: it fits a closed-form, per-head ridge mapper that (i) concatenates the top- k most predictive source layers for each target layer, (ii) strips RoPE from the keys before mapping so the fit is position-free and reusable across context lengths, and (iii)

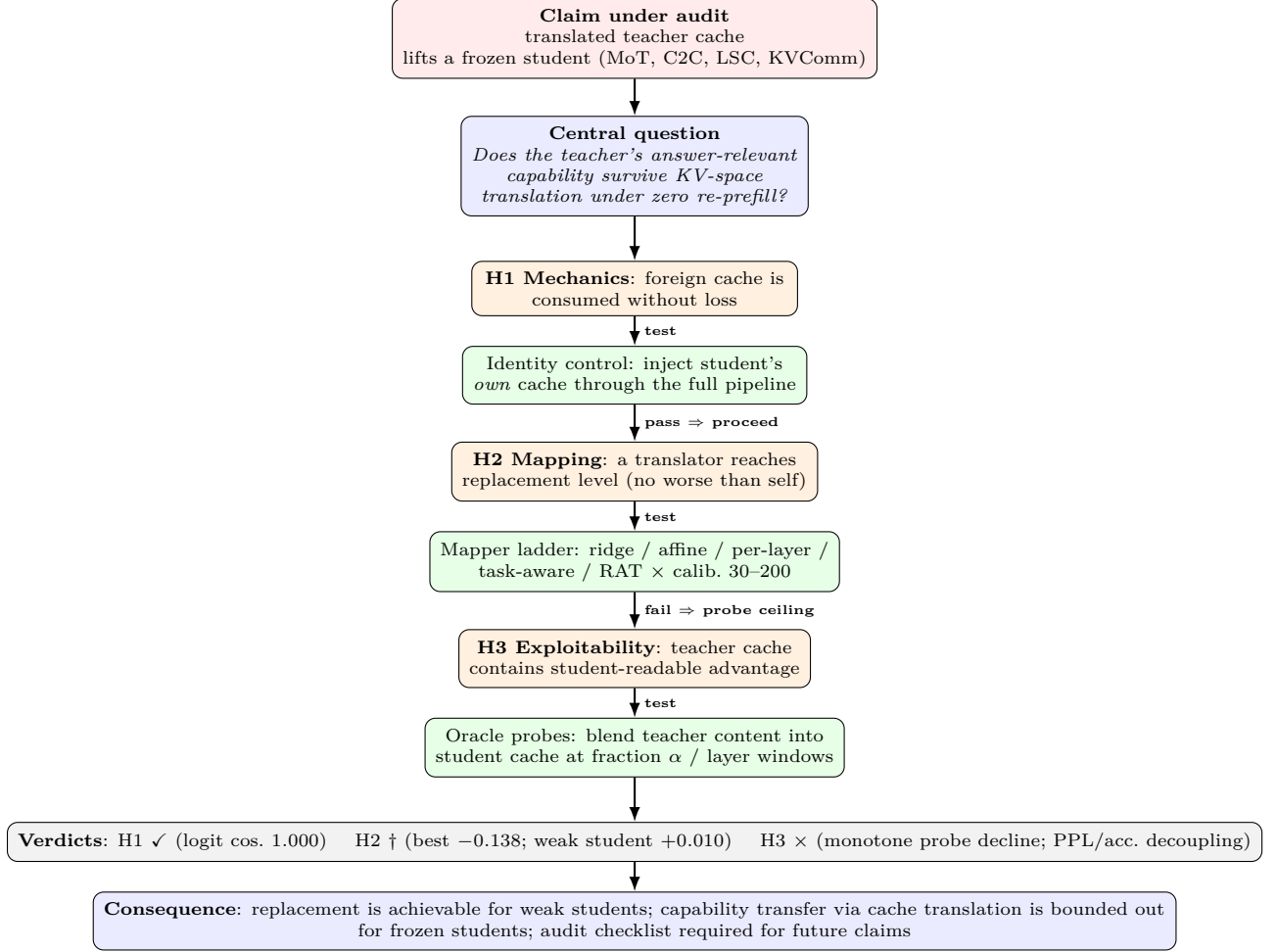


Figure 1: The three-hypothesis audit as a decision pipeline. Each hypothesis carries a dedicated instrument and a gate; H2’s failure routes the audit to H3’s oracle probes, which bound every translator without training one.

calibrates on 500 FineWeb-Edu sequences of 1,024 tokens each; across six matched-KV pairs in three families it retains 73–98% of the receiver’s standalone-prefill accuracy on four pairs while two degrade sharply, and an MLP recovers up to +37 pp HellaSwag retention on the failures [10]. KVComm shares attention-ranked KV layers [21]; Interlat communicates last-hidden-state messages [4]; HCache restores caches from hidden-state translations [6]. Latent Cache Flow translates sharer KV to a receiver through learned adapters and targets compact, bandwidth-efficient adapters [20]. Semantic Cache Distillation replaces raw KV with compact semantic codes: most layers are reconstructed by a low-rank cross-model projection, while a few key layers regenerate the receiver’s KV from the receiver’s own weights, framing state transfer as rate-distortion [17]. DroidSpeak shares KV caches between fine-tuned variants of one architecture by selectively recomputing a few layers and reusing the rest, so it does not address arbitrary large-to-small state synthesis [15]. A recent survey, Beyond Tokens, unifies these latent-communication lines and names cross-architecture alignment as open [14]. Two properties of this line matter for our audit. First, its evaluations score the translated pipeline against the student baseline, which conflates mechanics, mapping, and exploitability. The closed-form ridge mapper above is the exception on the prefill axis (its receiver skips prefill), but it still reports retention against a standalone prefill without isolating the three hypotheses. Second, the line’s other strongest variants do not operate under strict zero re-prefill: MoT reconstructs the target-side cache by replaying the context through the target model with source-guided sparse attention, so the reported quality includes a correction pass that a cold-start deployment would not get. Our audit holds the zero-re-prefill constraint fixed and asks what a

translator alone can deliver.

2.3 Cross-Model Representation Alignment

A separate line asks whether two trained networks’ representations can be aligned at all. Model stitching connects frozen networks with a learned linear layer and shows that stitching quality varies sharply across training runs and layers [1]. Relative representations replace absolute coordinates with anchor-based similarity spaces and enable zero-shot communication between encoders [18]. The Platonic Representation Hypothesis argues that representations converge across models up to linear transform [11], with refinements and proofs following [8, 25]. `vec2vec` translates embeddings between models without paired data by exploiting this shared geometry [12]. These results motivate the hypothesis that a linear map between cache spaces should exist. Our results qualify the conclusion for caches: the convergence evidence comes from pooled embedding spaces, while a cache is a per-layer, per-head, position-indexed state whose student-relevant content is not determined by its teacher counterpart (Section 6).

2.4 Frozen-Model Controllability

Work on steering shows that a frozen model’s behavior can be moved by injecting activations: task vectors compress in-context learning into a single mid-layer activation [9]; activation steering adds behavioral directions to the residual stream; gist tokens compress prompts into a few virtual tokens [19]. The tuned lens reads intermediate layers through per-layer affine probes [2], and the FFN-as-memory view assigns the factual capacity of a model to its feed-forward weights [7]. This line supports the intuition that a frozen student can absorb injected information, which makes the H3 null result informative rather than tautological: the failure is not that frozen students cannot be steered, but that the teacher’s cache does not carry the steering signal in student-readable form. It also suggests where transferable signal might live: in residual-stream directions and mid-layer task vectors, not in KV entries.

2.5 What the Survey Does Not Establish

Across the four lines, no work (i) separates mechanics from mapping from exploitability, (ii) injects the student’s own cache as an identity control, (iii) reports accuracy at matched perplexity, or (iv) bounds exploitability with an oracle that requires no training. The claims of the translation line therefore rest on pipelines whose components have never been isolated. The audit below supplies the missing instruments.

3 The Three-Hypothesis Audit Framework

3.1 Setup

Let a teacher T prefill a context x and form a cache $C_T(x) = \{K_\ell^T, V_\ell^T\}_{\ell=1}^{L_T}$ with L_T layers, H KV heads, and head dimension d . A translator g_θ maps it to $\widehat{C}_S(x)$ in the space of a student S with L_S layers. The student then answers a query q with only the query tokens as input, $\hat{y} = S(q \mid \widehat{C}_S(x))$, under strict zero re-prefill: the context never enters the student’s forward pass. Three references anchor the audit. *Student self-prefill* $S(q \mid C_S(x))$ is the baseline the handoff must not degrade. *Identity injection* feeds $C_S(x)$ itself through the injection pipeline; it isolates mechanics from mapping. *Teacher full* $T(q \mid C_T(x))$ is the capability upper bound. We measure answer quality by the gold-letter probability p_{gold} , accuracy, and the change $\text{CHG} = p_{\text{gold}}(\text{handoff}) - p_{\text{gold}}(\text{student self})$, with 95% bootstrap intervals and sign-flip permutation tests; perplexity over the scoring suffix is recorded but never accepted as evidence of capability (Section 5.5).

3.2 Hypotheses, Instruments, Gates

Figure 1 states the three hypotheses with their instruments and gates. **H1**: the identity injection matches the student’s own prefill per sample (gate: mean logit cosine = 1 within float tolerance). **H2**: some g_θ reaches replacement level (gate: $\text{CHG} > 0$ with CI excluding zero, across the mapper ladder at calibration budgets up to 200 examples). **H3**: the teacher’s cache contains student-readable advantage (gate: some oracle probe

Table 1: Cache-translation and adjacent work, organized by what is translated and what is held fixed. “Replay” marks methods whose reported quality includes a target-side correction pass over the context; our audit holds strict zero re-prefill. None of the cited evaluations uses an identity control or an exploitability probe.

Work	What moves	Alignment		Replay	Unexamined assumption
C2C [5]	Full KV	depth-ratio	layers	no	a linear map per channel suffices
LSC [3]	KV via shared latent	learned	latent space	no	one latent space fits all layers
MoT [13]	KV windows	gated translators		yes (sparse)	pipeline delta $\neq$ translation value
Heo et al. [10]	matched-KV full cache	per-head	ridge, de-RoPE	no	retention vs standalone; no identity control / probe
KVComm [21]	selected KV layers	attention	ranking	no	ranked layers carry the signal
Interlat [4]	last hidden state	latent message		no	the last layer summarizes the cache
HCache [6]	hidden states	translator		no	states determine caches
Latent Cache	KV via	learned adapter		no	compact adapter preserves receiver behavior
Flow [20]	adapters				rate-distortion suffices
SCD [17]	semantic codes (Reuse+Patch)	low-rank +	receiver weights	partial (Patch)	as proxy
DroidSpeak [15]	KV (same architecture)	selective	recompute + reuse	yes (selective)	fine-tuned variants share cache geometry
This paper	full cache (zero re-prefill)	five families incl. RAT		no	<i>none</i> : identity control + oracle bounds

configuration with teacher content improves over the student’s own cache). H3’s probe is decisive because it does not depend on training a translator; it directly measures what a frozen student can extract from teacher-cache content under the most favorable mixing.

4 Audit Methodology

4.1 Identity Control

At evaluation time we capture the student’s own cache over the context (allowed in calibration and probe stages), feed it through the same mapper interface, cache construction, position handling, and scoring path as a translated cache, and score. Any deviation from the student’s own prefill is mechanics loss. This is the control that prior evaluations omit.

4.2 Disjoint Splits, Unified Prompts, and Honest Counters

Mapper calibration and evaluation draws are disjoint samples of the same distribution (HellaSwag and ARC-Challenge, shuffled then split). All methods share one token-level prompt format, so the only difference between the handoff and the student baseline is where the context representation comes from. Zero-prefill counters read the actual cache length before and after the scoring forward pass rather than trusting book-keeping, and the scoring pass uses an isolated cache because the deployed transformers version mutates a cache even under `use_cache=False`, which silently pollutes co-located measurements.

4.3 Mapper Ladder

We audit five families under one interface. **Ridge (per-head)**: a 128×128 ridge map per student layer and head with Gram aggregation over variable-length calibration samples. **Affine (per-head)**: the same with mean centering and an intercept, fit in closed form over concatenated calibration samples. **Affine (per-layer)**: heads merged into rows, eight times fewer parameters. **Task-aware**: ridge initialization plus a bounded task fine-tune that mixes reconstruction with gold-answer cross-entropy (weight $\beta=0.3$) under a diagonal parameterization with a per-layer bounded gate. **RAT (residual-anchored translator)**: an architecture-derived map

$$W^{(s,h)} = [\pi(W_V^{S,h}) R \pi(W_V^{T,h})]^\top, \quad R = \arg \min_R \|E_T R - E_S\|_F^2 + \lambda \|R\|_F^2, \quad (1)$$

where $W_V^{T,h}$ and $W_V^{S,h}$ are the two models’ own value projections at the aligned layer pair, $\pi(\cdot)$ is the pseudo-inverse pullback from a head’s value space to the residual stream, and R aligns the two residual streams through the shared 151,936-token embedding matrices, which both Qwen3 models ship. A closed-form rank- r correction fits the calibration residual on top of the analytic core, a Hungarian matching on head covariance profiles replaces the identity head correspondence, and the sink position (position 0) is overwritten with the student’s calibration statistics because its translated value is a PPL catastrophe (Section 5.3). Keys follow the same construction on W_K . Every family is evaluated at calibration budgets 30 and 200 with disjoint evaluation samples ($n=30$ and $n=63$ respectively).

4.4 Oracle Probes

The probes assume an oracle that no training can beat: they hand the student its *own* cache and mix in teacher content at a controlled budget. **Fraction probe**: $\alpha C_S + (1 - \alpha) \hat{C}_S$ for $\alpha \in \{0.25, 0.5, 0.75\}$ with $\alpha=0$ the student’s own cache. **Window probe**: translated content in only the bottom, middle, or top third of student layers, student content elsewhere. Because every configuration keeps the student’s own cache dominant or localized, any improvement over $\alpha=0$ is a lower bound on what a perfect translator could deliver; monotone degradation bounds exploitability at zero. Probes are flagged non-deployable (they require the student’s own prefill) and exist only as measurement instruments.

4.5 Pairs, Data, and Protocol Discipline

Primary pair: Qwen3-4B (36 layers, hidden 2560) $\rightarrow$ Qwen3-1.7B (28 layers, hidden 2048); contrast pair: $\rightarrow$ Qwen3-0.6B (hidden 1024). Both share the same tokenizer, vocabulary, $H=8$ KV heads, $d=128$, and RoPE base 10^6 . Every run records its git commit and protocol version in its artifacts; 21 runs back the numbers below.

5 Results

5.1 H1 Passes: Mechanics Is Not the Failure Mode

The identity control reproduces the student’s own prefill exactly: per-sample logit cosine 1.000, gold CHG $+0.0001$ $[-0.0006, +0.0009]$, $n=30$. The injection pipeline, cache construction, GQA head handling, and position accounting are therefore correct, and every subsequent gap is attributable to the translation itself. The same control verified the disk-persistence path: a cold-start evaluation that loads a persisted cache and mapper parameters reproduces the offline run bit-for-bit.

5.2 H2 Fails for the Strong Student and Holds for the Weak One

Figure 2 and Table 2 give the landscape. Three regularities hold across all families. First, centering is the largest single gain: moving from per-head ridge (-0.296) to per-head affine (-0.138) halves the deficit, which confirms that the teacher/student value distributions differ by location and scale before they differ by content (the measured teacher/student norm ratios are $1.58\times$ for keys and $7.44\times$ for values). Second,

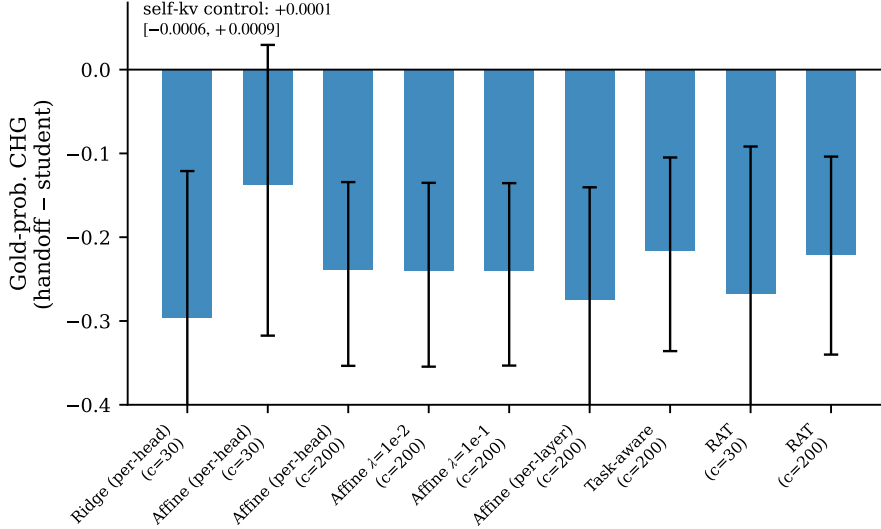


Figure 2: The mapper landscape for Qwen3-4B→1.7B under the unified audit protocol: gold-probability CHG over the student’s own prefill, with 95% bootstrap CIs. No family reaches zero; more calibration data moves affine mapping from -0.138 to -0.239 ; λ is inert over two orders of magnitude; the architecture-anchored RAT sits inside the same band.

the ladder is non-monotone in data: at 200 calibration examples the affine map degrades to -0.239 from -0.138 at 30, and λ over $[10^{-3}, 10^{-1}]$ changes nothing. The per-head linear hypothesis saturates; more data drives the estimator to the best linear approximation, which is worse than the strongly shrunk 30-example solution. Third, structure changes little: per-layer merging (-0.275), task-aware fine-tuning (-0.216), and RAT (-0.268 at $c=30$, -0.221 at $c=200$) sit inside the same band. The strong student never approaches its own prefill (0.503); the best mapper reaches 0.365. The weak 0.6B student tells the opposite story: affine at $c=200$ gives $+0.010$ $[-0.081, +0.102]$, statistically indistinguishable from its own prefill, which is replacement-level service for a weak student.

5.3 RAT Component Ablations Separate Fluency From Capability

RAT ablations attribute the failure inside the architecture-anchored design. Removing the sink override explodes suffix perplexity to 2.7×10^5 : the translated position-0 entry, which carries outsized attention mass, is a catastrophe unless overwritten with the student’s own statistics. Removing the analytic core and raising the correction rank to 32 *repairs fluency*: PPL drops to 23.7, within 2.5 of the identity control. Accuracy stays at 0.267. The analytic core’s min-norm pullback is harmful on real models, and more importantly, fluency recovery does not touch the answer deficit. The cache can be made to look right to the language-modeling objective without becoming right for the task.

5.4 H3: Oracle Probes Bound Exploitability at Zero

The probes answer H3 without training anything (Figure 3). With 75% student cache and 25% teacher content, gold probability already falls from 0.5031 to 0.3798; the decay is monotone through $\alpha=0.75$. Window injection localizes the damage: replacing the top third of layers is indistinguishable from the student (0.5022 versus 0.5031), while the middle and bottom thirds are strongly harmful (0.2653, 0.2242). The reading is architectural. Answer-relevant routing runs through early and mid layers, exactly where the teacher’s cache is least compatible; the top layers, where the translated content is least damaging, are not where the answer is decided. Under the most favorable conditions any translator could hope for, the teacher’s cache adds nothing and subtracts much. Exploitability is bounded at zero.

Table 2: Main audit results. Acc/gold/PPL: answer accuracy, gold-letter probability, and perplexity of the scoring suffix for the translated cache (kv-both). CHG is the gold-probability change over the student’s own prefill with 95% bootstrap CIs. Pairs: Qwen3-4B→1.7B unless noted. All runs use disjoint calibration/evaluation splits and the unified prompt format.

Configuration	Acc / Gold / PPL	Gold CHG [95% CI]
Identity control (self-kv), 1.7B	0.500 / 0.503 / 21.2	+0.000 [−0.001, +0.001]
Student self-prefill, 1.7B	0.500 / 0.503 / 25.4	–
Teacher (full prefill), 1.7B	0.767 / 0.693 / –	–
Native average, 1.7B	0.167 / 0.169 / 536082.5	−0.334 [−0.508, −0.154]
Ridge per-head, c=30, 1.7B	0.200 / 0.207 / 88.5	−0.296 [−0.479, −0.121]
Affine per-head, c=30, 1.7B	0.367 / 0.365 / 46.1	−0.138 [−0.317, +0.029]
Affine per-head, c=200, 1.7B	0.254 / 0.258 / 56.4	−0.239 [−0.354, −0.134]
Affine per-layer, c=200, 1.7B	0.206 / 0.222 / 81441.1	−0.275 [−0.407, −0.140]
Task-aware, c=200, 1.7B	0.270 / 0.280 / 142.1	−0.216 [−0.336, −0.105]
RAT, c=30, 1.7B	0.233 / 0.235 / 501.6	−0.268 [−0.440, −0.092]
RAT, c=200, 1.7B	0.238 / 0.276 / 1008.2	−0.221 [−0.340, −0.104]
RAT no-anchor rank-32, c=30, 1.7B	0.267 / 0.280 / 23.7	−0.223 [−0.367, −0.088]
Affine per-head, c=200, 0.6B	0.302 / 0.283 / 71.8	+0.010 [−0.081, +0.102]
RAT, c=200, 0.6B	0.206 / 0.232 / 790.0	−0.041 [−0.120, +0.038]

5.5 Perplexity and Accuracy Decouple

Figure 4 plots perplexity against accuracy for every method and run. The identity control and the student occupy the top-left regime. Translated caches populate a broad band, including a point with near-native perplexity and a 23-point accuracy deficit. Native injection sits at extreme perplexity (5.4×10^5) with low accuracy. Perplexity measures whether the cache permits fluent continuation; accuracy measures whether the cache supports the task. The two dissociate, so perplexity cannot serve as evidence of translation quality, a practice that appears in the serving literature because perplexity is cheap.

5.6 Channel Asymmetry

Channel ablations at $c=30$ rank value-only (0.367) above key-only (0.300) above both (0.200) on accuracy, while perplexity orders them in reverse (88 versus 253 versus 1.8×10^7). Mapping keys repairs fluency and damages routing: the mapped keys steer attention to plausible but task-irrelevant positions. We report the asymmetry as a finding and leave the mechanism open.

6 Why Translation Fails: An Architectural Analysis

The audit’s verdicts fit the architecture. A cache entry is a linear projection $W_V h$ of a residual-stream state h . Translation must recover h from V , move it across the residual-space gap (2560 to 2048), and re-project. Two facts bound the recovery. First, the pullback $\pi(W_V)V$ returns only the row-space component of h ; the component orthogonal to W_V ’s rows, which can dominate a 2560-dimensional state observed through a 128-dimensional head, is invisible to the teacher’s own cache and therefore unavailable to any translator, however nonlinear. The student’s corresponding value $W_V^S R h$ depends on that invisible component, so V_S is not a function of V_T : the mapping task is intrinsically underdetermined, and a learned translator can at best track the conditional mean. Second, the teacher’s advantage over the student lives in weights, not in caches: FFN layers carry the model’s factual associations [7], and the capability gap between Qwen3-4B and 1.7B is a gap in these parameters, which a cache does not transport. The oracle probes make the consequence concrete: even with the student’s own cache holding 75% of the mass, the teacher’s 25% contribution degrades the answer. MoT’s own design corroborates the analysis: its reported quality includes a target-side replay with a correction loss, which is exactly the upper-layer correction that strict zero re-prefill denies. Under a cold start, the translator’s output is the whole story, and the whole story does not contain the teacher’s capability.

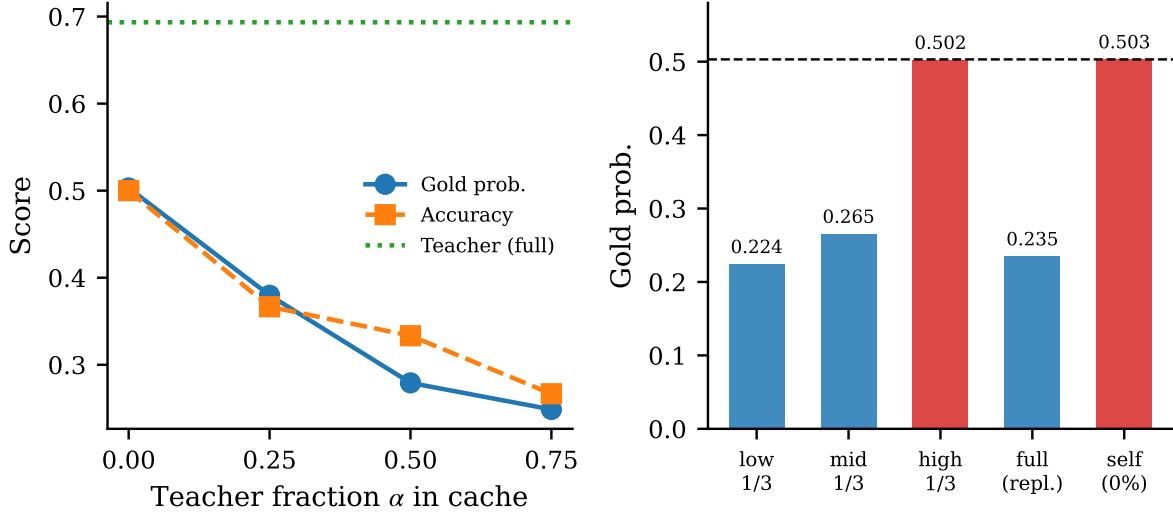


Figure 3: Oracle probes on Qwen3-4B→1.7B. Left: blending teacher-cache content into the student’s own cache degrades gold probability and accuracy monotonically in the teacher fraction α ; $\alpha=0$ is the student’s own cache. Right: window injection is harmless only in the top third of layers; full replacement and early/mid injection are far below the student.

7 Implications for Cache-Translation Research

For claims. Any cache-translation evaluation should ship an identity control, a gold-probability metric with intervals, and an oracle probe. A pipeline delta over a student baseline is not evidence of translation; a perplexity gain is not evidence of capability. Our instruments are one evaluation run each and compose with any serving stack.

For deployment. Two regimes survive the audit. For weak students relative to the teacher (here 0.6B under 4B), translated injection reaches parity with self-prefill, so cached handoff saves the student’s prefill at no measured quality cost. Top-layer injection is harmless; if a system must spend translation capacity, the top layers are where it can be spent, though our probes also show that spending it there buys nothing. For strong students, cache translation is currently a net negative, and the honest alternative is the text channel: the teacher’s summary of the context scores 0.413 against the student’s 0.508 full-text baseline, so compression loses less than translation loses on the strong student.

For the direction. The survivable versions of the idea relocate work from the cache to weights or to the replay: allow a target-side correction pass as MoT does, train student-side adapters, or translate in a shared latent space anchored by task vectors rather than per-entry caches. Each option trades away part of the cold-start guarantee, and each should be re-audited with the identity control before its delta is attributed to translation.

8 Limitations

Our audit covers one model family (Qwen3), two pairs, and two multiple-choice tasks with $n=30$ –63 evaluation samples per run; the CI widths reflect that. The oracle probe bounds exploitability under linear blending and three-layer windows; a nonlinear exploit path outside this family would not be detected, though it would also need to survive the same measurement protocol. RAT’s analytic core assumes value projections are informative pullbacks of the residual stream; the synthetic validation confirms the mathematics, and the real-model failure is itself an audit finding. We did not re-implement MoT’s full replay pipeline; our claims concern the zero-re-prefill regime that its predecessors and our deployment target share. All runs use one hardware and one seed per configuration; the multi-seed sweep in the appendix-scale runs showed the same signs.

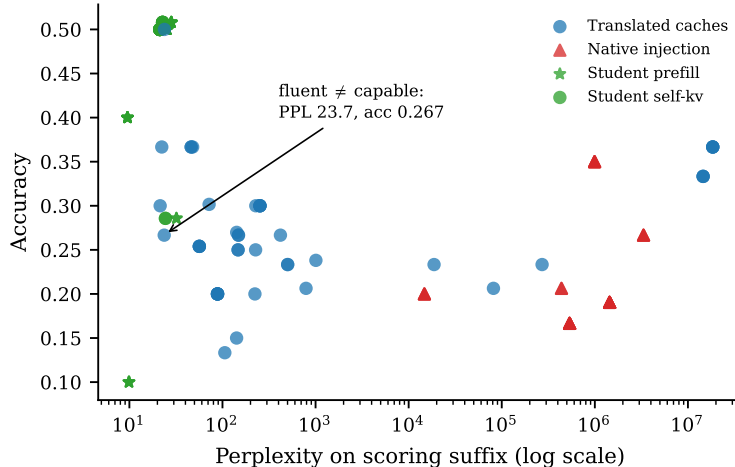


Figure 4: Perplexity over the scoring suffix versus accuracy across all methods and runs (log scale). The identity control and the student sit at low PPL and high accuracy; translated caches span two regimes; the best-reconstructed cache (PPL 23.7) remains 23 accuracy points below the student.

9 Conclusion

Cache translation promises to move a teacher’s work into a smaller model. The audit separates that promise into three testable parts. The mechanism works: an identity control reproduces the student’s own prefill exactly. The map does not: five mapper families, a calibration ladder, and an architecture-anchored translator all leave the strong student below its own prefill, and the weak student at parity. The reason is not the map: oracle probes bound the student-readable advantage of the teacher’s cache at zero, and a near-native-perplexity cache still fails the task. The teacher’s capability is in its weights. We release the audit framework so that the next cache-translation claim can be checked in one run.

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A Reproducibility

All numbers come from the `apcs` repository’s recorded runs (`reports/runs/*/inject-eval/`), each carrying its git commit, protocol version, configuration, and per-sample scores. The audit protocol is implemented in `apcs.inference.evaluator` with the flags `calib_eval_split`, `calib_shuffle`, `probe_mode`, and `native_rescale`; mapper families live in `apcs.mapper` (`math`, `joint`, `task_aware`, `rat`); persistence and the cold-start online phase in `apcs.inference.kv_store`. Unit tests cover the audit counters, the store round-trip, and the mapper interface.

B Additional Measurements

The teacher/student KV norm ratios are $1.58\times$ (keys) and $7.44\times$ (values), measured over 30 calibration contexts per layer. Norm rescaling of the native cache repairs neither its perplexity (5.4×10^5) nor its accuracy, ruling out scale as the primary failure. The unified prompt renders choices once; the earlier protocol rendered them twice for the student baseline, which inflated the confidence-based deltas reported in our own preliminary sweeps and motivated the unified format in the audit.